Joshua Dimasaka

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SKILLS

PROGRAMMING

Proficient:
MATLAB • QGIS• Google Earth Engine
Experienced:
Python • R • Stata • Tableau • LaTeX•

RESEARCH

Climate & Disaster Risk Assessment
Earthquake Risk & Loss Modelling
Deep Probabilsitic Modelling
Geospatial Data Science
Spatiotemporal Analysis

OPEN-SOURCE PROJECTS

- VBCI [★ 14] [66 cites] [242]
  Joint inference of multi-hazards & building damage from satellite images by causal dependency modelling (*Nature Communications*)
- METEOR 2.5D: 72-GB Data [960]
  Spatiotemporal maps of physical vulnerability in Least Developed Countries, 1975-2030 (*AAAI'26*)
- DeepC4 [2k]
  Deep conditional census-constrained clustering for multi-task spatial disaggregation of urban morphology (*AGU'24, ISPRS P&RS, in review*)

EDUCATION

UNIVERSITY OF CAMBRIDGE

PH.D. ARTIFICIAL INTELLIGENCE FOR THE STUDY OF ENVIRONMENTAL RISKS
Oct. 2023 -- Sep. 2026 | Cambridge, UK
Dissertation: Understanding Regional Dynamic Physical Risks, Advanced Geospatiotemporal Modelling of Exposure and Vulnerability Using Deep Constrained Clustering and Probabilistic Temporal Graph Learning with Earth Observation Data

UNIVERSITY OF CAMBRIDGE

M.RES. ENVIRONMENTAL DATA SCIENCE
Oct. 2022 -- Sep. 2023 | Cambridge, UK

STANFORD UNIVERSITY

M.A. PUBLIC POLICY
Mar. 2021 -- Jun. 2022 | Stanford, CA, USA

STANFORD UNIVERSITY

M.S. CIVIL & ENVIRONMENTAL ENG'G
Sep. 2019 -- Jun. 2022 | Stanford, CA, USA

STANFORD UNIVERSITY GRADUATE SCHOOL OF BUSINESS

EXECUTIVE EDUCATION CERTIFICATE IN ENTREPRENEURSHIP & INNOVATION
Mar. 2021 -- Jun. 2022 | Stanford, CA, USA

UNIVERSITY OF THE PHILIPPINES

B.S. CIVIL ENGINEERING
Jun. 2013 -- Jan. 2018 | Los Baños, PHL
magna cum laude, college valedictorian (1/341)
completed one-semester early

EXPERIENCE

GERMAN AEROSPACE CENTER (DLR) | VISITING RESEARCHER

Jul. - Sep. 2024 | Apr. - Jun. 2026 | Munich, Germany | 6 months

- Led development of a geospatial ML framework & benchmark dataset for building vulnerability & change detection using satellite embeddings & census-based constraints.

EARTHQUAKES & MEGACITIES INITIATIVE (EMI) | URBAN RESILIENCE FELLOW

Mar. - Aug. 2022 | Quezon City, Philippines | 5 months

- Developed large-scale seismic risk assessment tools covering 400,000 buildings & 3.2M population for a scenario "Big One" M7.2 earthquake.

STANFORD LAND, BUILDINGS, & REAL ESTATE (LBRE) | SEISMIC RISK CONSULTANT

Mar. 2021 - Mar. 2022 | Stanford, CA, USA | 1 year

- Quantified earthquake-induced financial losses and business interruption risks for 750 buildings at Stanford University under seismic scenarios up to magnitude 7.6, and developed a Tableau dashboard for stakeholder decision-making.

FM GLOBAL, ENG'G & RESEARCH | STRUCTURES & NATURAL HAZARDS RESEARCHER

Jun. - Sep. 2021 | Norwood, MA, USA | 4 months

- Improved the firm's library of earthquake design hazard and maps by more than 246%, starting with 130 and ending with 320 maps (or 190 new maps) from over 130 countries.

STANFORD UNIVERSITY, STRUCTURES AS SENSORS RESEARCH LAB | GEOSPATIAL & MACHINE LEARNING GRADUATE RESEARCHER

Jun. 2020 - Jun. 2021 | Stanford, CA, USA | 1 year

- Implemented a Bayesian causal inference framework leveraging satellite imagery to enhance landslide, liquefaction, & structural damage models across the USA.

SOUTH BAY CONSULTING ENGINEERS | STRUCTURAL ENGINEERING INTERN

Jun. - Sep. 2020 | San Jose, CA, USA | 3 months

- Produced structural designs and resolved review comments while leading retrofit of a 9,000 ft^2 two-story home with a weak first story using a cantilevered column system.

NEW STORY CHARITY, RESEARCH & DEV'T TEAM | HOUSING RESEARCH FELLOW

Feb. - Jun. 2020 | San Francisco, CA, USA | 4 months

- Evaluated 7 affordable housing material innovations and led feasibility studies with 20 interdisciplinary fellows across finance, policy, architecture, engineering, and construction.

ARUP | STRUCTURAL ENGINEER

Jul. 2018 - Jul. 2019 | Metro Manila, Philippines | 1 year

- Conducted post-earthquake assessment, performance- and code-based structural peer reviews for multiple 41–50 story high-rise developments, and performed liquefaction susceptibility assessment for a three-tower project in a reclaimed land area.

MAKATI DEVELOPMENT CORPORATION | PLANNING & CONTROL ENG'G INTERN

May. - Jul. 2017 | Bonifacio Global City, Philippines | 3 months

- Built a VBA-Excel system tracking deliverables across 100+ concurrent projects.

PUBLICATIONS

Nature Communications, AAAI (AI for Social Impact), Environmental Research Letters, ISPRS Photogrammetry & Remote Sensing (*in review*), Progress in Disaster Science (*in review*)

SERVICE

2025	Lead Researcher, *Mw-7.7 Earthquake Impact Assessment in Myanmar & Thailand*
2017	Lead Organizer, *12th Engineering Congress (Philippine, National)*
2014-22	Lead Coordinator & Interviewer, *Ramar Scholarship Foundation*

AWARDS

2026	Digital GreenTalents Award, *High Potentials in Sustainable Dev't (20/global)*
2024	Helmholtz Researcher Grant ($15k), *Helmholtz Association, Germany*
2022-26	UKRI CDT Studentship ($290k), *Eng'g & Physical Sciences Research Council, UK*
2019-22	Knight-Hennessy Scholars ($240k), *Stanford University, USA (69/4,424)*
2018	BPI-DOST Science Award, *Bank of the Philippine Islands, PHL*